# AMIN MEHRVARZ

+1 (617) 758-9882 ◇ amin.mehrvarz@gmail.com ◇ Boston, MA, US, 02210
Linkedin ◇ Google Scholar

## EDUCATION

**Ph.D. Mechanical Engineering - Robotics/Mechatronics** ***Sep 2017 - Aug 2021***
Northeastern University, Boston, MA, U.S **Overall GPA**: 3.87/4
**Dissertation**: Theoretical and Experimental Analyses of Tunable Active Metamaterials via Use of Feed-forward/back Controllers

**M.Sc. Mechanical Engineering - Control** ***Sep 2013 - Sep 2015***
Sharif University of Technology, Tehran, Iran **Overall GPA**: 3.82/4
**Thesis**: Positioning, tracking and vibrating boundary control in micro-beams with considering piezo-electric actuators effects.

**B.Sc. Mechanical Engineering - Design and Manufacturing** ***Sep 2008 - Aug 2013***
Sharif University of Technology, Tehran, Iran **Overall GPA**: 3.8/4
**Thesis**: Design and manufacture of a mold placement mechanism for a hydraulic press.

**Diploma, Mathematics and Physics** ***Sep 2004 - Jun 2008***
Dehkhoda, Karaj, Alborz, Iran **Overall GPA**: 4/4

## RESEARCH INTERESTS

| | | | |
|---|---|---|---|
| Control Theory, | Robotics, | Mechatronics, | Machine Learning |
| Multibody Dynamics, | Wave Propagation, | Vibrations | Acoustics |

## SELECTED COURSES

| | | |
|---|---|---|
| Robotics, | Advanced Control, | Advanced Vibrations |
| Mechatronics, | Linear Control, | Modal Analysis |
| Digital Signal Processing, | Nonlinear Control, | Vibrations of continues systems |
| Advanced Dynamics, | Adaptive Control, | Advanced Mechanisms Design |
| Machine Learning, | Optimal Control, | Advanced Mathematics |

## COMPUTER SKILL

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|---|---|---|---|---|---|
| MATLAB, | Python, | C++, | Simulink, | Simscape, | ROS2, |
| COMSOL, | Maple, | ANSYS, | SolidWorks, | CATIA, | AutoCAD |

## EXPERIENCE

**Medtronic, Boston, MA, U.S.** **Jan 2022–Present**
*Sr. Software Engineer–Developer, Surgical Robotics* Sep 2024–Present

- Project Lead: next-gen fully-wristed Tri-Staple integration into the Hugo software platform.
- Delivered the first software prototype with Unity simulation on the Hugo surgeon console for surgeon evaluation.
- Developed the control and kinematic software for the Tri-Staple surgical robot stapler instrument.

- train-Gauge Drop-Out Estimator: inferred torque & velocity from real sensor data at 98% accuracy for Tri-Staple.
- Validated the drop-out estimator via Monte Carlo sampling across learning-time and drop-out parameters.
- Real-Time Inverse Kinematics via Jacobian-Based Optimization: eliminated singularities for smoother haptic feedback.
- Developed low-jerk trajectory generators and closedloop motion controllers in MATLAB/Simulink for robotic surgical stapler instruments, enabling smooth position, velocity, and acceleration tracking.
- Designed and integrated real-time embedded control software for precision-driven robotic manipulators.
- Conducted hardware-in-loop testing, ensuring seamless integration between control software and medical devices.
- Collaborated with cross-functional teams (systems engineering, software, and validation) to define and review safety-critical software requirements.
- Reviewed MATLAB/Simulink models, verifying compliance with precise standards for medical device software quality.
- Participated in design reviews and contributed to the documentation of architecture, unit testing, and anomaly tracking.
- Integrated mechanical, electrical, and software components, particularly in an embedded, high-performance control environment.

*Sr. Software Engineer–Test Lead, Surgical Robotics* Jun 2023–Sep 2024

- Test Lead for software release in surgical robot and leading software verification for FDA submission
- Provide technical leadership for the software verification engineering verifying the functionality of feedback control loop algorithms designed and developed in MATLAB/Simulink
- Lead a team verifying controls software for robotic manipulators and surgical instruments
- Lead a team integrating and testing new controller designs on hardware and with the rest of the robotic platform
- Conduct software and product hardware-in-loop testing and create model and prototypes
- Review teammates MATLAB/Simulink models and ensure that all software meets quality standards; prepare and participate in design reviews
- Document or participate in the documentation of the software including verification test plans and reports
- Lead the definition and implementation of different software testing techniques to increase test coverage in both automation and manual testing.
- Understand the technical behavior of the system, architecture, software requirements with an aptitude to identify product defects in all phases of development.
- Lead development of solutions that have significant improvements of processes and test development with aptitude for continue improvement.

- Work with the Project Management Office to coordinate dry runs and formal verification activities as required
- Collaborate with technical engineering teams, mentor junior test engineers.

*Controls Verification Engineer, Surgical Robotics* Jan 2022–Jun 2023

- Conducted software and producted hardware-in-loop testing and created model and prototypes
- Reviewed MATLAB/Simulink models to ensure that all software meets quality standards
- Documented and participated in the documentation of the software including verification test plans and reports
- Implemented different software testing techniques in Python and Matlab to increase test coverage in both automation and manual testing.
- Analyzed the technical behavior of the system, architecture, software requirements with an aptitude to identify product defects in all phases of development.
- Led development of solutions that have significant improvements of processes and development with aptitude for continue improvement.
- Worked with the Project Management Office to coordinate dry runs and formal verification activities as required
- Collaborated with technical engineering teams, mentor junior test engineers.
- Developed the verification test methods to evaluate performance metrics of complex, electromechanical medical systems

**Northeastern University, Boston, MA, U.S** **Sep 2017–Dec 2021**

*Graduate Research Assistant* *Sep 2017–Dec 2021*

- Conducted doctoral research in robotics, control systems, and mechatronics, with an emphasis on dynamic modeling, feedback control, and tunable active metamaterials.
- Developed and analyzed mathematical models and control strategies using MATLAB and Simulink to investigate the dynamic behavior and performance of complex mechanical systems.
- Applied theoretical analysis, numerical simulation, and experimental validation to evaluate control methods and system performance.
- Disseminated research findings through peer-reviewed publications and conference presentations, contributing to 10 journal articles and 7 conference papers.
- Served as an instructor for the following courses: Advanced Dynamics, Advanced vibration, Mechatronics, and Measurement Laboratory.
- Coordinated the activities and operations of the Piezoactive Systems Laboratory (PSL).

**Sharif University of Technology, Tehran, Iran** **Sep 2013 – Sep 2015**

*Graduate Research Assistant* *Sep 2013 – Sep 2015*

- Conducted research in nonlinear control systems, partial differential equation (PDE)-based control, adaptive control, and fluidstructure interaction.
- Applied mathematical modeling, numerical simulation, and computational methods to investigate dynamic systems, vibration suppression, and control system stability.
- Published 3 journal articles and 1 conference paper based on research findings.

- Delivered instruction in undergraduate engineering courses, including Automatic Control, Dynamics of Machinery, Vibrations, Dynamics and Vibrations Laboratory, and Mechanism Design.

**Power Planet Repairs, Internship, Tehran, Iran** **Jun 2013 – Aug 2013**

- Modeled and redesigned an 8000kg gas turbine with Catia and rebuilt the model in Solidworks
- Simulated the model of the gas turbine by SolidWorks and ANSYS with different loads
- Analyzed the damages on the system and determined the effects of the cracks on the system's failure
- Presented a SolidWorks workshop to improve the knowledge of the company's staff

**Sharif University of Technology, Internship, Tehran, Iran** **Jun 2011 – Aug 2011**

- Research on different available FEA software including ANSYS, Fluent, SolidWorks's modules & Comsol Multiphysics
- Modified an available solenoid valve and documented the steps of it
- Calculated and Analyzed the solenoid valve flow rate using SolidWorksFlow simulation
- Compared the results of the modified valve with the original one, and showed performance improvement by

## PUBLICATIONS

Google Scholar

- Mehrvarz, A., Khodaei, M. J., Darabi, A., Zareei, A., & Jalili, N. (2024). Experimental demonstration of broadband reconfigurable mechanical nonreciprocity. Journal of Vibration and Control, 30(1-2), 288-294.
- Khodaei, M. J., Mehrvarz, A., Ghaffarivardavagh, R., & Jalili, N. (2022). Acoustic meta-silencer: Toward enabling ultrawide-band air-permeable noise barrier. Journal of Vibration and Control, 28(15-16), 1928-1936.
- Khodaei, M. J., Mehrvarz, A., Ghaffarivardavagh, R., & Jalili, N. (2022). Retrieving effective acoustic impedance and refractive index for size mismatch samples. AIP Advances, 12(6).
- Korivand, S., Mehrvarz, A., Candelino, N., Khodaei, M. J., & Jalili, N. (2021). Band Gap and Stiffness Tuning of a Sandwich Plate by Magnetostrictive Materials. ASME Letters in Dynamic Systems and Control, 1(4), 041007.
- Mehrvarz, A., Khodaei, M. J., Clark, W., & Jalili, N. (2020, November). A new dynamic model of a two-wheeled two-flexible-beam inverted pendulum robot. In ASME International Mechanical Engineering Congress and Exposition (Vol. 84553, p. V07BT07A044). American Society of Mechanical Engineers.
- Khodaei, M. J., Mehrvarz, A., Ghaffarivardavagh, R., & Jalili, N. (2020). Non-centric cavity-based acoustic metasurface: Enabling full phase modulation. AIP Advances, 10(9), 095020.
- Khodaei, M. J., Candelino, N., Mehrvarz, A., & Jalili, N. (2020). Physiological closed-loop control (PCLC) systems: Review of a modern frontier in automation. Ieee Access, 8, 23965-24005.
- Salehzadeh, R., Candelino, N., Khodaei, M. J., Mehrvarz, A., & Jalili, N. (2020, October). Parameter Sensitivity Analysis of Piezoelectrically-Actuated Flexural/Torsional Vibrating Beams. In Dynamic Systems and Control Conference (Vol. 84287, p. V002T38A008). American Society of Mechanical Engineers.

- Khodaei, M. J., Inaloo, M. H. B., Mehrvarz, A., & Jalili, N. (2019). An Adaptive Neuro-fuzzy Strategy in Closed-loop Control of Anesthesia. arXiv preprint arXiv:1910.06325.
- Ardekany, A. N., Daneshmand, F., & Mehrvarz, A. (2019). Vibration Analysis of a Micropolar Membrane in Contact with Fluid. Iranian Journal of Science and Technology, Transactions of Mechanical Engineering, 43(1), 695-704.
- Mehrvarz, A., Najafi Ardekani, A., Khodaei, M. J., & Jalili, N. (2019). Vibration analysis and control of fluid containers using piezoelectrically-excited side wall. Journal of Vibration and Control, 25(7), 1393-1408.
- Khodaei, M. J., Mehrvarz, A., & Jalili, N. (2019). An adaptive multi-critic neuro-fuzzy control framework for intravenous anesthesia administration. IFAC-PapersOnLine, 51(34), 202-207.
- Mehrvarz, A., Salarieh, H., Alasty, A., & Vatankhah, R. (2018). Boundary vibration control of strain gradient timoshenko micro-cantilevers using piezoelectric actuators. arXiv preprint arXiv:1812.01155.
- Khodaei, M. J., Mehrvarz, A., Candelino, N., & Jalili, N. (2018, September). Theoretical and experimental analysis of coupled flexural-torsional vibrations of rotating beams. In Dynamic Systems and Control Conference (Vol. 51913, p. V003T42A004). American Society of Mechanical Engineers.
- Mehrvarz, A., Khodaei, M. J., Clark, W., & Jalili, N. (2018, September). Modeling and dynamics analysis of a beam-hoverboard self-transportation system. In Dynamic Systems and Control Conference (Vol. 51913, p. V003T32A008). American Society of Mechanical Engineers.
- Mehrvarz, A., Salarieh, H., Alasty, A., & Vatankhah, R. (2018). Vibration boundary control of Timoshenko micro-cantilever using piezoelectric actuators. Scientia Iranica, 25(2), 711-720.
- Ardekany, A. N., & Mehrvarz, A. (2016, October). Vibrations suppression of a Euler-Bernoulli beam in contact with a fluid using piezoelectric actuators. In 2016 4th International Conference on Robotics and Mechatronics (ICROM) (pp. 285-288). IEEE.

## PROJECTS

**Control of Active Metamaterial**

- Proposed and designed a feedback/forward controller to spatiotemporal modulate the metamaterials stiffness using MATLAB
- onfigured and analyzed wave propagation in the controlled system with %15 variable stiffness using MATLAB/COMSOL
- Studied and validated wave propagation experimentally using piezo sensors/actuators, hardware-in-the-loop, Simulink & dSPACE
- Analyzed the real-time data and investigated the nonreciprocity feature of the controlled system using MATLAB/Simulink

**Closed-Loop Control of Anesthesia**

- Proposed and advanced a medical device controller with %20 drug consumption reduction in anesthesia using Python
- Optimized and trained the adaptive multi-critic neuro-fuzzy controllers coefficients using Python
- Designed an optimal and adaptive controller for automation in anesthesia with no overshoot using Python

**Moving Robot**

- Analyzed and compared PID and MPC controllers effects on the stability of a flexible robot using MATLAB
- Developed and designed arm robot gripper to detect different types of fish using machine learning/Python

**Control of active baffles**

- Innovated and introduced a smart fluid tank to eliminate the %99 of fluid movements within 0.01s using FEA/MATLAB
- Presented a robust nonlinear controller to stabilize the fluid movement using feedback linearization techniques
- Compared the designed feedback controller with the available products and showed the %30 more efficiencies of it using MATLAB

**Acoustics wave manipulation**

- Developed and initiated a metasurface for complete phase modulation using MATLAB/COMSOL
- Presented and investigated a methodology for broad-band sound attenuators in air-permeable structures
- Developed a theory-simulation method to measure the acoustic properties of metamaterials using MATLAB/COMSOL

**PDE controllers for Atomic Force Microscopy (AFM)**

- Examined and compared the effects of different PDE controllers on AFM using C++
- Presented and simulated an ODE feedback controller to eliminate the vibrations of AFM in 0.001s using C++
- Introduced and implemented a Lyapunov-based boundary controller to %99 stabilize the movement of AFM using C++

**Control of Coronary Artery Model**

- Designed and evaluated chaos controller to reduce the error of coronary artery system to %1 using MATLAB/Simulink
- Developed a sliding mode model reference, adaptive controller, for the coronary artery's nonlinear model using Matlab/Simulink

**Fluid-Structure Vibrations**

- Investigated and analyzed the vibrations of micropolar membranes in contact with fluid using C++
- Monitored and interpreted rotor-shafts behavior under the influence of fluid forces using MATLAB

## PROCEEDINGS CONFERENCES

- International Mechanical Engineering Congress and Exposition, ASME 2020, Virtual, USA
- Dynamic Systems and Control Conference, ASME 2020, Virtual, USA
- Learning for Dynamics and Control (L4DC), 2019, Cambridge, MA, USA
- 2nd IFAC Conference on Cyber-Physical and Human Systems CPHS 2018, Miami, Florida, USA
- Dynamic Systems and Control Conference, ASME 2018, Atlanta, Georgia, USA

- 4th International Conference on Robotics and Mechatronics (ICROM), IEEE 2016, Tehran, Iran

## LANGUAGE

Persion (Native)  English (Advanced)  Arabic (Beginner)

## HONNORS AND AWARDS

- Awarded Outstanding Graduate Researcher, College of Engineering, Northeastern University, 2020
- Awarded Outstanding Teaching, College of Engineering, Northeastern University, 2021
- Awarded Outstanding Reviewer, IOP Publishing, 2020
- Awarded scholarship, Mechanical and Industrial Engineering Dept, Northeastern University, 2017
- Ranked 14th out of 20,000 participants, Nationwide entrance exam for M.Sc. degree, 2013
- Ranked 196th out of 350,000 participants, Nationwide Entrance exam for B.Sc. degree, 2008

## EXTRACURRICULAR ACTIVITIES

- American Society of Mechanical Engineers (ASME) Member
- International Federation of Automatic Control (IFAC) Member

## REVIEWER EXPERIENCE

- Reviewer of Smart Materials and Structure Journal
- Reviewer of IEEE Access Journal
- Reviewer of Journal of Vibration and Control
- Reviewer of 2nd IFAC Conference on Cyber-Physical and Human-Systems 2018
- Reviewer of the 4th RSI International Conference on Robotics and Mechatronics (ICROM)